Si5351A VFO Rev A — Field Manual Ch. 6. Plan: PLAN.md (2026-10-22). Hierarchy: power/USB, MCU/UI, synth/RF.
Inter-sheet nets via global labels (MCU_TX, MCU_RX, DTR, I2C_SCL, I2C_SDA) and power rails.

Power and USB

File: power_usb.kicad_sch

MCU and UI

File: mcu_ui.kicad_sch

Synth and RF outputs

File: synth_rf.kicad_sch

Draft schematic, not reviewed. See VERIFY notes.

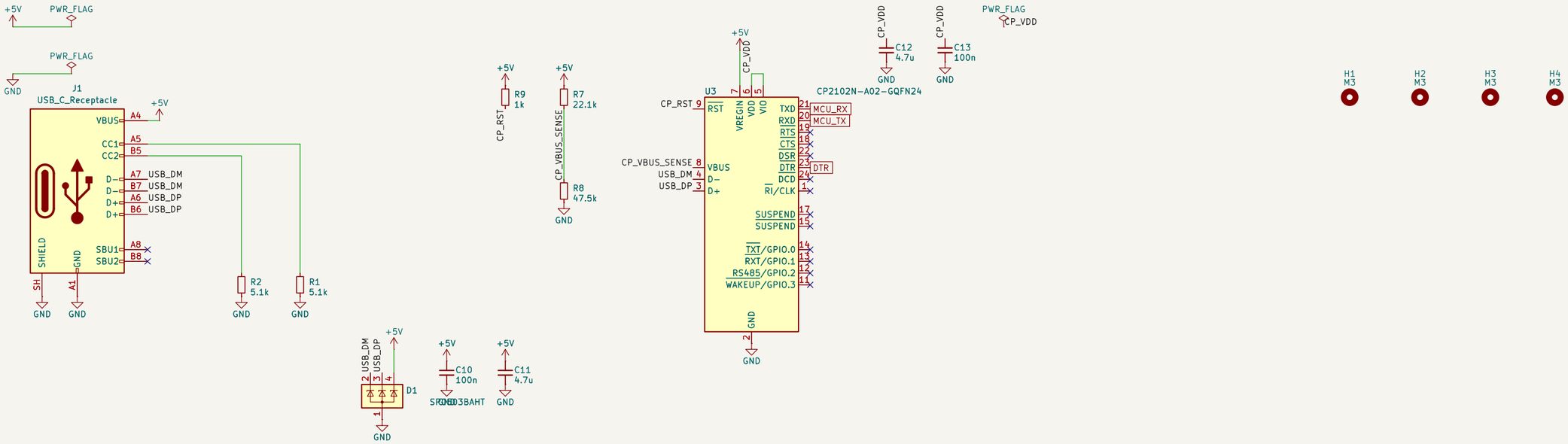
Etherkit / NT7S

Sheet: /
File: Si5351AVFO.kicad_sch

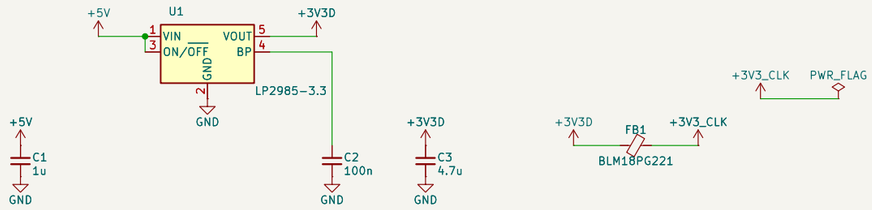
Title: Si5351A VFO — Top

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/4

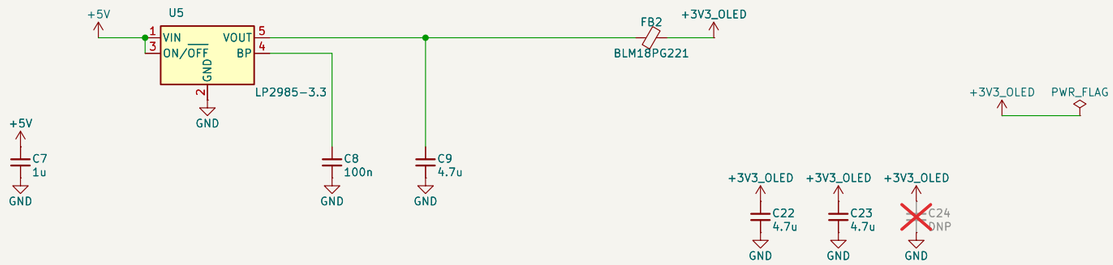
USB-C input, CP2102N USB-UART (from EtherkitUARTAdapter), LP2985 rails (U1 from Rev F; U5 dedicated to OLED per Jason 2026-10-22)



U1: +3V3D (MCU) and, via FB1, +3V3_CLK (Si5351 + TCXO). Rev F U1/C1/C2/C3.



U5: OLED-only rail (Eyes On The Prize lesson, PLAN 4a). Place U5/FB2/C22-C24 at the OLED header.



VERIFY (unconfirmed items on this sheet):
- J1: UART adapter schematic pairs GCT USB4105 footprint with this HCTL LCSC part; HCTL footprint used here — confirm. Shield to GND assumed (not stated in PLAN).
- D1: Pin-to-line mapping (D-/D+/VBUS on pins 2/3/4) to be checked against adapter layout.
- C10: LCSC stock low; fallback C49678 (0805) per PLAN
- U3: Unused CP2102N pins left NC (adapter used DTR via JP1, TRRS dropped). Confirm none were strapped on the adapter.
- R9: PLAN says /RST pull-up to VBUS (as on adapter); CP2102N RST max is VIO-referenced — confirm adapter really ties to VBUS.
- FB2: Bead or series R (<=6 ohm upper bound, PLAN 4a); value to be set by measurement.
- C22: OLED bulk value: PLAN start point 2x4.7u, tune by measurement
- C24: DNP 1206 site for larger OLED bulk cap; value/part not chosen

Draft schematic, not reviewed. See VERIFY notes.

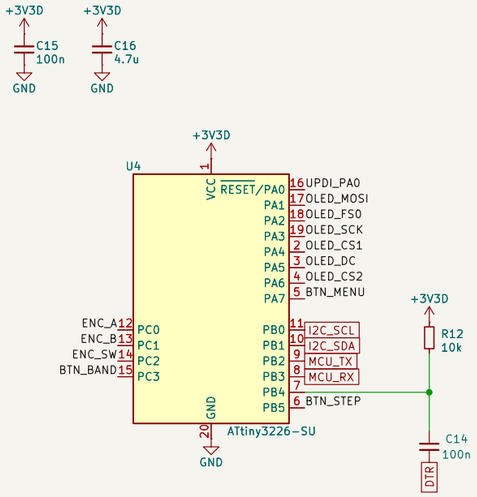
Etherkit / NT7S

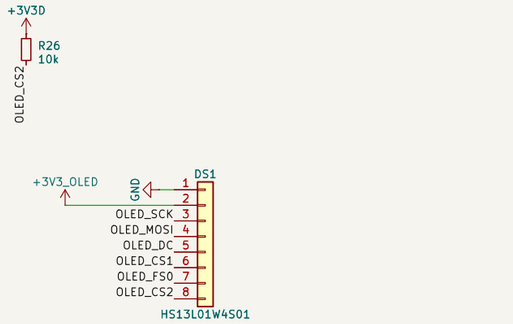
Sheet: /Power and USB/
File: power_usb.kicad_sch

Title: Si5351A VFO — Power and USB

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/4

ATtiny3226 @ 8 MHz internal, 3.3 V. Optiboot over CP2102N; DTR auto-reset on PB4 (RSTPINCFG=0x3, megaTinyCore "UPDI w/reset on PB4"), as EtherKeyerMini C3/R4.

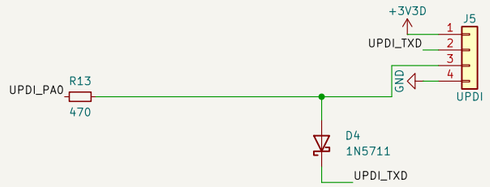


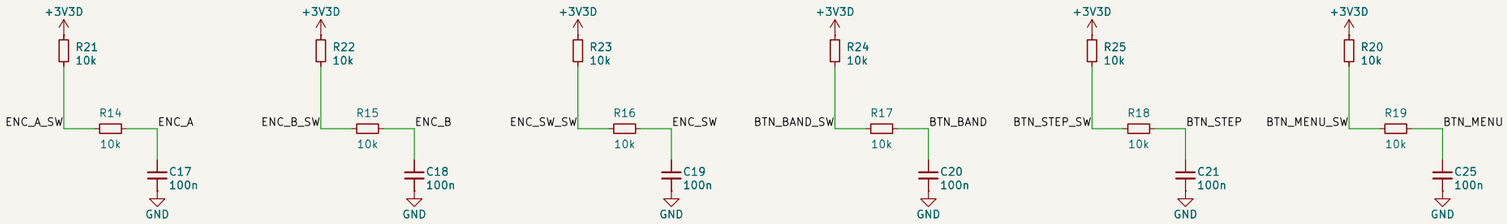


OLED: SPI only (PA1 MOSI, PA3 SCK, PA4 CS1, PA5 DC). Never shares the Si5351 I2C bus.
CS2 (font chip) held high by R26.

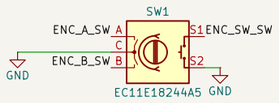
UPDI / serial-UPDI header, circuit copied from EtherKeyerMini (J3,R3,D1 -> J5,R13,D4;
EKM D2/JP2/JP3 omitted per Jason). Pin 1 VCC ref (+3V3D), 2 TXD, 3 RXD/UPDI, 4 GND.

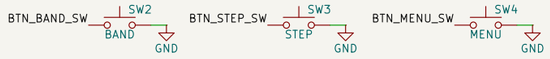
Pin 1 = VCC reference for programmer; board must be USB-powered while programming.





VERIFY (unconfirmed items on this sheet):
- U4: Pin mux (PLAN sec.3) to be checked vs datasheet I/O mux table. Jason consigning parts.
- C14: PLAN: series R or clamp on PB4 for DTR-rising overshoot? EKM has none.
- J5: Replicated from EtherKeyerMini (J3); EKM has no LCSC/MPN fields — part number to choose.
- R13: Replicated from EtherKeyerMini (R3); EKM has no LCSC/MPN fields — part number to choose.
- D4: Replicated from EtherKeyerMini (D1); EKM has no LCSC/MPN fields — part number to choose.
- DS1: 1.3in SPI SH1106 module. Pin order GND,VCC,SCL,SDA,DC,CS1,FS0,CS2 from PLAN pin list; pin numbering, pitch and connector style unverified. Low stock — Jason sourcing.
- C17: Debounce topology/count: PLAN lists x5 but 6 inputs (A,B,SW,3 buttons) — drawn as 6. Check vs encoder bounce spec.
- SW1: Detent/pulse count unverified.





Draft schematic, not reviewed. See VERIFY notes.

Etherkit / NT7S

Sheet: /MCU and UI/
File: mcu_ui.kicad_sch

Title: Si5351A VFO — MCU and UI

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/4

Si5351A + 25 MHz TCXO copied from Si5351A Breakout Rev F (R5/C5/C6/R6/C4, XB open). Each output: Rs 0R -> Cb 100n DC block -> DNP shunt -> LPF (DNP) with 0R bypass -> SMA.

VERIFY (unconfirmed items on this sheet):
- X1: TCXO stock thin (718); buy before ordering.
- R27: I2C pull-up value not fixed (PLAN 2.2-4.7k); no LCSC chosen. On +3V3D: ATtiny, SI5351, CP2102N and OLED all at 3.3 V, so I2C levels match.
- R28: as R27
- R100: 0603 0R series (Jason decision); LCSC code not in PLAN (C17477 is 0805).
- C100: DC-block LF corner ~32 kHz into 50 ohm (PLAN Q6).
- R101: DNP shunt ~120 ohm back-match of ~85 ohm source (datasheet Zo); part/LCSC not chosen; verify return loss esp. >100 MHz.
- L1: LPF inductor size (SMD vs T37 option, PLAN Q7)
- C105: Parallel C across L (elliptic) per LowPassFilterModule - unconfirmed
- C101: LPF values builder choice (DNP)
- J2: Footprint is a placeholder; check BWSMA-KE-P001 launch geometry and 1.6 mm edge fit.
- R120: see R100
- R121: see R100
- J3: see J2
- R140: see R100
- R141: see R100
- J4: see J2

+3V3_CLK
C4
100n
GND

+3V3_CLK
R5
10
PWR_FLAG
R6
0
C5
100n
GND
X1
25MHz TCXO
NC/EN
VDD
GND
OUT
C6
100n
GND

+3V3_CLK
U2
XA
XB
VDD
VDDO
CLK0
CLK1
CLK2
GND
I2C_SCL
I2C_SDA
Si5351A-B-GT
GND

+3V3D
R27
VERIFY 2.2k-4.7k
I2C_SCL

+3V3D
R28
VERIFY 2.2k-4.7k
I2C_SDA

Bypass: RC (R104/R124/R144) fitted by default; RA/RB DNP. Filter build: fit RA/RB, remove RC.
Keep RA/RB pads <=2 mm from junction (PLAN 6).

CLK0 output chain (J2)

CLK0
R100 0
C100 100n
R101 120
R102 0
C101 DNP
C102 DNP
C103 DNP
C104 DNP
R103 0
L1 DNP
L2 DNP
L3 DNP
C105 DNP
C106 DNP
C107 DNP
R104 0
J2
SMA CLK0
GND

CLK1 output chain (J3)

CLK1
R120 0
C120 100n
R121 120
R122 0
C121 DNP
C122 DNP
C123 DNP
C124 DNP
R123 0
L4 DNP
L5 DNP
L6 DNP
C125 DNP
C126 DNP
C127 DNP
R124 0
J3
SMA CLK1
GND

CLK2 output chain (J4)

CLK2
R140 0
C140 100n
R141 120
R142 0
C141 DNP
C142 DNP
C143 DNP
C144 DNP
R143 0
L7 DNP
L8 DNP
L9 DNP
C145 DNP
C146 DNP
C147 DNP
R144 0
J4
SMA CLK2
GND

Draft schematic, not reviewed. See VERIFY notes.
Etherkit / NT7S
Sheet: /Synth and RF outputs/
File: synth_rf.kicad_sch
Title: Si5351A VFO - Synth and RF outputs
Size: A3 | Date: 2026-10-22 | **Rev: A**
KiCad E.D.A. 10.0.7 | Id: 4/4